Molecular Cell, Volume *70*

Supplemental Information

Sequence, Structure, and Context Preferences of Human RNA Binding Proteins

Daniel Dominguez, Peter Freese, Maria S. Alexis, Amanda Su, Myles Hochman, Tsultrim Palden, Cassandra Bazile, Nicole J. Lambert, Eric L. Van Nostrand, Gabriel A. Pratt, Gene W. Yeo, Brenton R. Graveley, and Christopher B. Burge

Figure S1. RBNS assay and comparison to RNAcompete, Related to Figure 1.
A. Histogram of the number of unique reads sequenced in all RBNS Input and Pulldown libraries, with the ~250,000 RNAcompete oligo sequences marked for comparison.
B. 10th-90th percentile range of the mean base-pairing probability (P_{paired}) for all 5-mer occurrences (averaged over all five positions) in RBNS input libraries and the RNAcompete custom array. The 1,024 5-mers are sorted along the *x*-axis by increasing median RBNS P_{paired}.
C. Dendrogram of clustered RBNS (red) and RNAcompete (black) experiments for the 31 RBPs that were assayed by both methods. Clustering was performed via the hclust function in R (with method="average") on the pairwise Pearson correlations of 5-mer *R* Z-scores, for the set of 306 5-mers that were enriched (Z≥3) in any of the 62 experiments.
D. Histogram of Pearson correlations of RBNS and RNAcompete for the 31 RBPs assayed by both methods as described in **C**.

Fig. S1

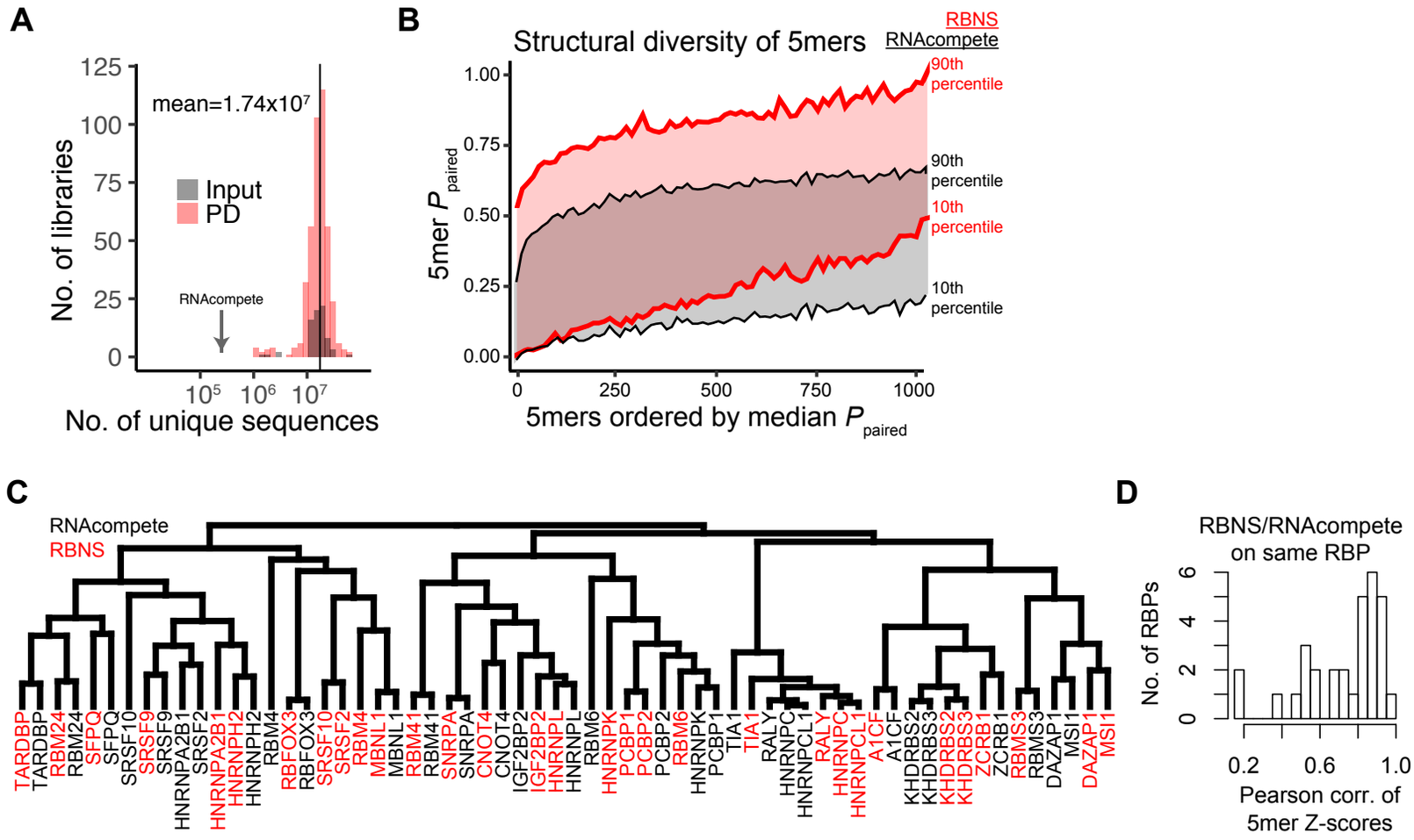

A
mean=1.74x10^7
Input
PD
RNAcompete
No. of libraries
No. of unique sequences
B
Structural diversity of 5mers
RBNS
RNAcompete
90th percentile
90th percentile
10th percentile
10th percentile
5mer P paired
5mers ordered by median P paired
C
RNAcompete
RBNS
TARDBP
TARDBP
RBM24
RBM24
SFPQ
SFPQ
SRSF10
SRSF9
SRSF9
HNRNPA2B1
SRSF2
HNRNPA2B1
HNRNPH2
HNRNPH2
RBM4
RBFOX3
RBFOX3
SRSF10
SRSF2
RBM4
MBNL1
MBNL1
RBM41
RBM41
SNRPA
SNRPA
CNOT4
CNOT4
IGF2BP2
IGF2BP2
HNRNPL
HNRNPL
RBM6
HNRNPK
PCBP1
PCBP2
PCBP2
RBM6
HNRNPK
PCBP1
TIA1
TIA1
RALY
HNRNPC
HNRNPCL1
RALY
HNRNPC
HNRNPCL1
A1CF
A1CF
KHDRBS2
KHDRBS3
KHDRBS2
KHDRBS3
ZCRB1
ZCRB1
RBMS3
RBMS3
DAZAP1
MSI1
DAZAP1
MSI1
D
RBNS/RNAcompete on same RBP
No. of RBPs
Pearson corr. of 5mer Z-scores

Figure S2. Overlapping specificities of RBPs, Related to Figure 2.
A. Number of unique top 6-mers among random subsamplings of the RBNS experiments versus randomly selected 6-mers (similar to **Fig. 2C**), for the subset of 78 RBNS experiments that excludes any paralogs (left) or any RBPs that share at least 40% identity among any RBDs (right).
B. Similar to **Fig. 2C** and **Fig. S2A**, but for the top 15 6-mers of each RBNS experiment instead of just the top 6-mer. Black line determined from sets of 15 6-mers in which the top 6-mer was chosen at random, with the remaining 14 6-mers matching the edit distances relative to the top 6-mer observed among actual RBNS experiments (**Methods**).
C. Mapping of the four nucleotide frequencies in motif logos onto a 2D simplex, for both the actual 78 RBNS motifs (left), simulated RBNS motifs (center), and the resulting enrichment of RBNS versus simulated motifs in each simplex partition (right) (**Methods**). Gray boxes denote that none of the 78 RBNS motif frequencies mapped to that partition. Significance along margins of the enrichment simplex was determined by bootstrap Z-score (number of asterisks = Z-score). The data in the top and bottom rows is the same, with the A and C corners of the simplex switched so that each of the six dinucleotide combinations (AC, AG, AU, CG, CU, GU) is included along an edge in at least one of the two mappings. Upper right simplex is the same as **Fig. 2F**.

Fig. S2

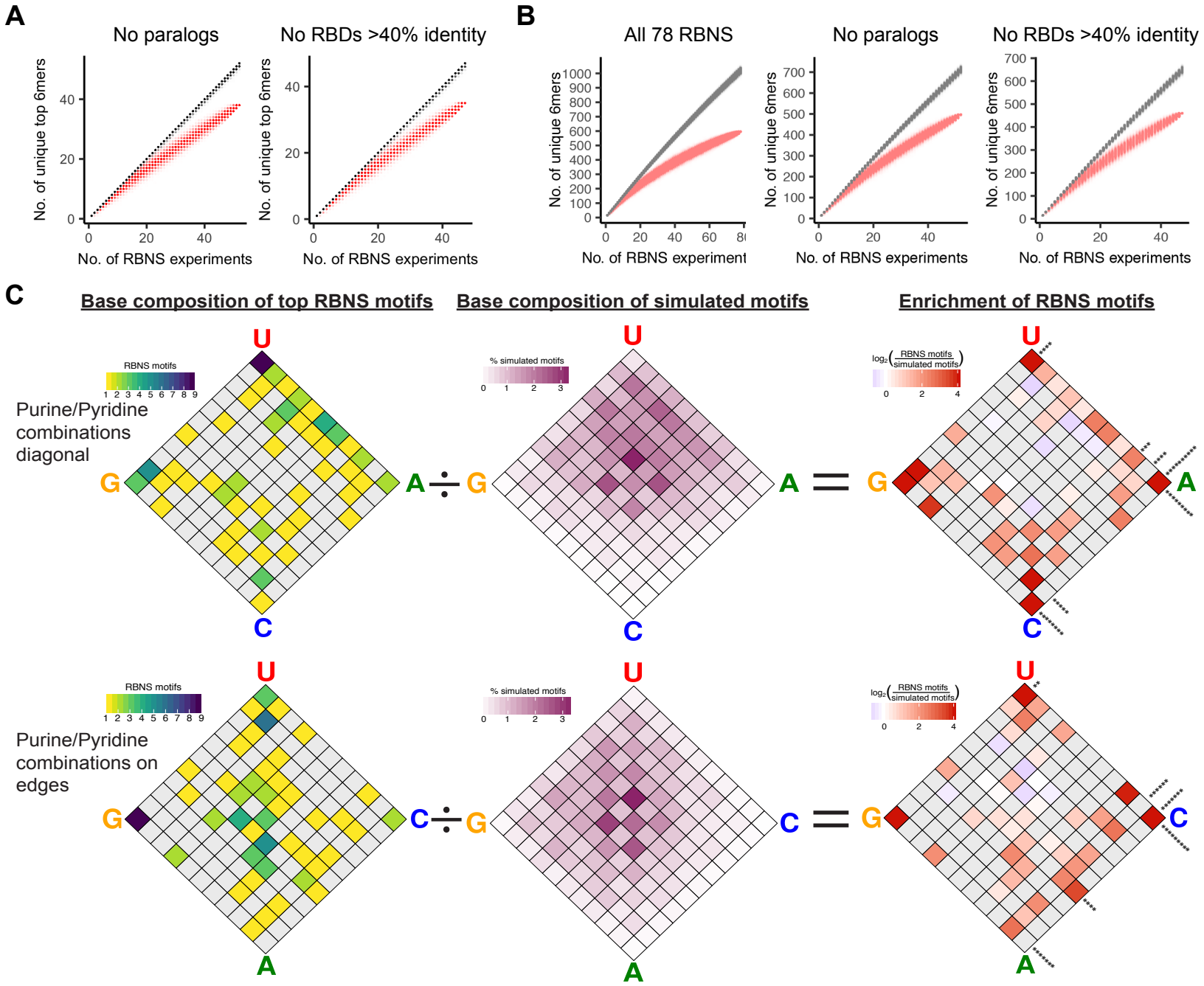

A
No paralogs
No RBDs >40% identity
No. of unique top 6mers
No. of RBNS experiments
B
All 78 RBNS
No paralogs
No RBDs >40% identity
No. of unique 6mers
No. of RBNS experiments
C
Base composition of top RBNS motifs
Base composition of simulated motifs
Enrichment of RBNS motifs
RBNS motifs
% simulated motifs
log2(RBNS motifs / simulated motifs)
Purine/Pyridine combinations diagonal
Purine/Pyridine combinations on edges
U
G
A
C

Figure S3. RBNS-derived splicing and stability RNA maps and RBP binding in the transcriptome, Related to Figure 3.

A. Summary of RBNS-inferred regulatory activity of RBPs in pre-mRNA splicing and mRNA levels. **Left (Splicing)**: From left to right: All RBPs assayed by both RBNS and KD/RNA-seq, ordered from top to bottom by RBNS motif (same ordering as in **Fig. 2A**); a bar denoting KD in HepG2/K562; the top RBNS motif logo as in **Fig. 2A**; a bar denoting whether there were significantly more SEs included upon KD (pink) or excluded upon KD (blue); the inferred RBNS regulatory activity of the RBP over the SE and upstream and downstream 250nt of flanking intron (**Methods**), with strength of RBNS splicing regulatory activity of significant regions noted by green (ESE/ISE) or red (ESS/ISS) heat map; total number of SEs changing in each direction upon KD. **Right (mRNA levels)**: Similar to Left (Splicing), but for inferred activity on gene expression levels based on RBNS motif density in the 3' UTRs of genes significantly up- or down-regulated upon KD (Destab.=increased RBNS density in 3' UTRs of genes up- regulated upon KD; Stab.=increased RBNS density in 3' UTRs of genes down-regulated upon KD).

B. Similar to **Fig. 3F** right, but for SRSF5 KD in K562 (left) and HepG2 (right) cells.

C. Comparison of RBNS and eCLIP motifs for RBPs assayed by both techniques (reproduced with permission from (Van Nostrand et al., 2017)). The top RBNS and eCLIP motifs are shown for each RBP (Methods), clustered by RBNS motifs as in **Fig. 2A**. 17 of the 24 RBPs with significant overlap in the 5-mers comprising the RBNS and eCLIP logos ($P<0.05$, hypergeometric test) marked with a star to the right of the eCLIP logo.

D. Left: Enrichment of RBNS 5-mers (averaged among all RBNS 5-mers with Z-score≥3) in eCLIP peaks that occur in both replicates (peaks overlap by at least 1 position) relative to peaks that occur in only one replicate. Right: Enrichment of RBNS 5-mers (averaged among all RBNS 5-mers with Z-score≥3) in eCLIP peaks that occur in both HepG2 and K562 (peaks overlap by at least 1 position) relative to peaks that occur in only one cell type. Error bars are boostrapped standard deviations (100 samples).

E. Genome browser snapshot of a portion of the 3' UTR of the *STOM* gene and eCLIP entropies ($\log_2$(IP density/input density)) for TIA1 and HNRNPC bound to two different locations in this region. Instances of U_5 motifs shown in top track (red) and eCLIP peaks are marked in black below each entropy track. Motifs derived from RBNS and eCLIP peaks for each RBP shown on the left.

Fig S3

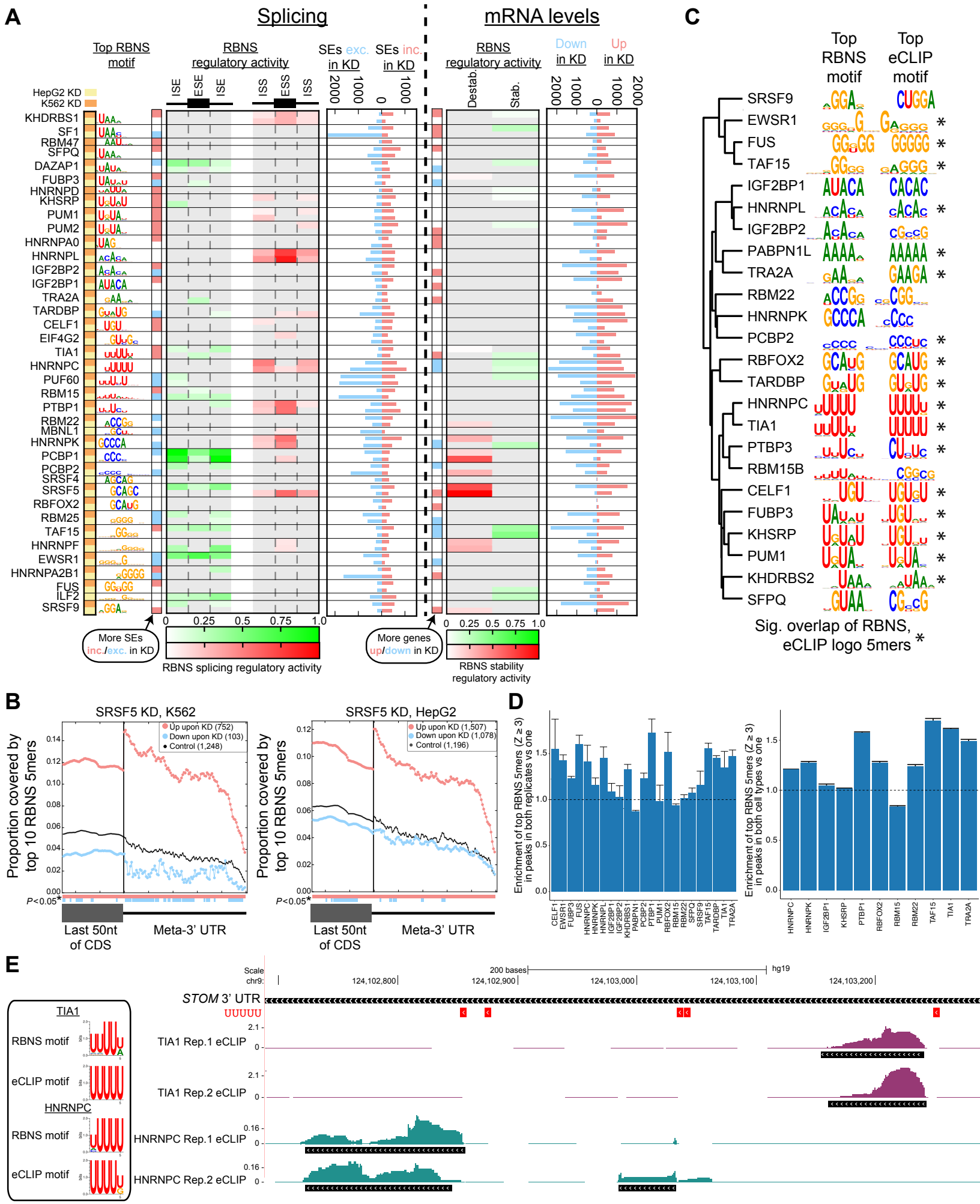

A
Splicing
mRNA levels
Top RBNS motif
RBNS regulatory activity
SEs exc. in KD
SEs inc. in KD
RBNS regulatory activity
Down in KD
Up in KD
HepG2 KD
K562 KD
ISE
ESE
ISE
ISS
ESS
ISS
Destab.
Stab.
More SEs inc./exc. in KD
RBNS splicing regulatory activity
More genes up/down in KD
RBNS stability regulatory activity
C
Top RBNS motif
Top eCLIP motif
Sig. overlap of RBNS, eCLIP logo 5mers *
B
SRSF5 KD, K562
SRSF5 KD, HepG2
Proportion covered by top 10 RBNS 5mers
Up upon KD (752)
Down upon KD (103)
Control (1,248)
Up upon KD (1,507)
Down upon KD (1,078)
Control (1,196)
P<0.05*
Last 50nt of CDS
Meta-3' UTR
D
Enrichment of top RBNS 5mers (Z ≥ 3) in peaks in both replicates vs one
Enrichment of top RBNS 5mers (Z ≥ 3) in peaks in both cell types vs one
E
TIA1
RBNS motif
eCLIP motif
HNRNPC
STOM 3' UTR
TIA1 Rep.1 eCLIP
TIA1 Rep.2 eCLIP
HNRNPC Rep.1 eCLIP
HNRNPC Rep.2 eCLIP
200 bases
hg19
chr9:
124,102,800
124,102,900
124,103,000
124,103,100
124,103,200

Figure S4. *In vitro* and *in vivo* structural preferences of RBPs and distribution of enrichments across reads, Related to Figure 4.

A. Top: P_{paired} over each position averaged over the 78 RBPs; 95% confidence interval is shadowed. Bottom: Mean P_{paired} in the most enriched pulldown library over the top 6-mer plus 10 flanking positions on each side; RBPs are grouped by motif clusters in **Fig. 2A**. Right: Mean change ($\log_2$) in pulldown vs. input P_{paired} averaged over the top 6-mer.

B. Correlation of *R* value profiles of the top 5-mer across five bins of increasing P_{paired} for RBPs assayed by both RBNS (x-axis) and eCLIP (y-axis). For each assay, *R* was calculated in each of the 5 P_{paired} bins (as in **Fig. 4C**) and the monotonicity of *R* over the 5 bins was calculated (-10 monotonicity = 5 bins monotonically decreasing *R* with increasing P_{paired}; 10 monotonicity = 5 bins monotonically increasing *R* with increasing P_{paired}).

C. The frequency of the top RBNS 6-mer at each position of the random region, relative to a uniform distribution at all positions (top: RBPs with random 20-mers; bottom: random 40-mers). Difference from uniform denoted by the green heat map bar to the right, calculated as the KL-divergence of the observed frequency at each position relative to a uniform distribution; RBPs are sorted by decreasing difference. RBPs and their top 6-mers noted on left, with ZNF326 and RBM22 marked as the 1st and 3rd most unequal distributions across 20-mers, respectively.

D. Quantification of filter binding assay performed on ZNF326 with A) an oligo containing the RBNS motif and structure identified in **Fig. 4F**, B) an oligo that preserves the structure while disrupting the motif, C) an oligo that preserves the motif while disrupting the structure, and D) random RNA (N_{12}). Errors bars represent the max and min across replicate experiments.

Fig. S4

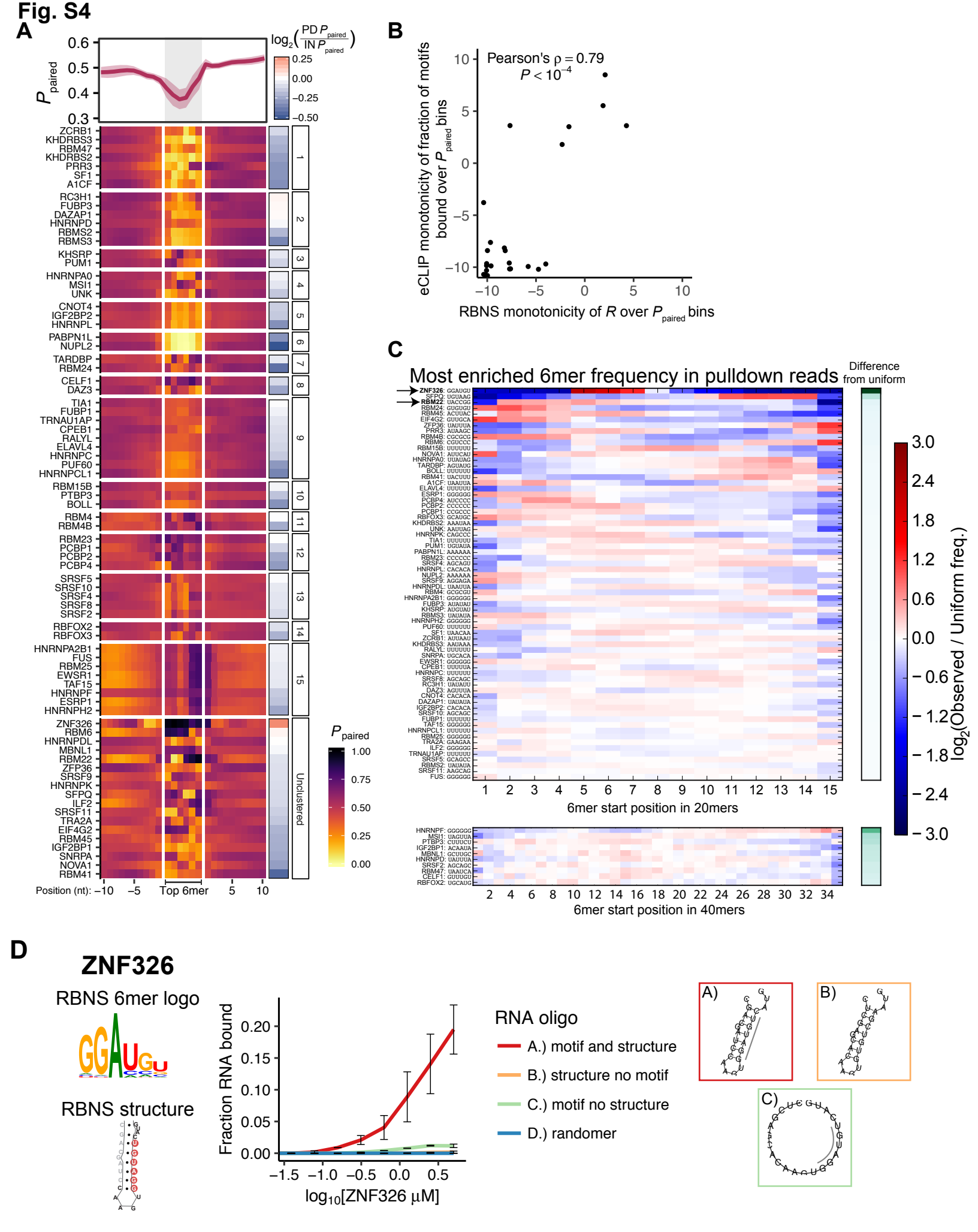

A
B
C
D
Pearson's ρ = 0.79
P < 10^−4
eCLIP monotonicity of fraction of motifs bound over P_paired bins
RBNS monotonicity of R over P_paired bins
Most enriched 6mer frequency in pulldown reads
Difference from uniform
6mer start position in 20mers
6mer start position in 40mers
log2(Observed / Uniform freq.)
Position (nt): −10 −5 Top 6mer 5 10
Unclustered
ZNF326
RBNS 6mer logo
RBNS structure
Fraction RNA bound
log10[ZNF326 μM]
RNA oligo
A.) motif and structure
B.) structure no motif
C.) motif no structure
D.) randomer

Figure S5. Bipartite core spacing, flanking nucleotide composition, and degenerate pattern binding preferences, Related to Figure 5.

A. Quantification of filter binding assays done on DAZAP1 (top) and RBM45 (bottom) with the most enriched linear 6-mer and the most enriched split 6-mer. Error bars represent max and min across replicate experiments.

B. Enrichment ($\log_2$) as a function of the spacing between pairs of cores (3-mers) for NOVA1 (top), UNK (middle), and PTBP3 (bottom). Individual 3-mer pairs are shown in blue, and the average across all pairs is shown in gray (analogous to **Fig. 5A, B**).

C. Relative preference for spacing (**Methods**) grouped by whether RBPs have a single RBD, multiple RBDs, or have been shown to multimerize in the literature with the domain responsible for multimerization included in the RBNS construct. *P*-value determined by *t*- test.

D. Flanking nucleotide compositional preferences for RBPs with KH domains, with and without controls for RNA secondary structure. "Matched P_{paired}" contains motif occurrences that are sampled to have the same mean base-pairing probability in the input and pulldown libraries. "Motifs in hairpins" contains motif occurrences that have the 5-letter code "HHHHH" in the minimum free energy structure in both the input and pulldown libraries.

E. Enrichment for particular nucleotides in reads with no high-affinity motifs for FUS (left) and IGF2BP1 (right).

F. Enrichments for degenerate patterns of length 12 with 6 fixed bases shown for HNRNPK (left) and RBFOX2 (right). Only the top 5% of degenerate patterns are shown and the red lines indicate the enrichment of the linear 6-mers. For HNRNPK, degenerate patterns where 5 out of the 6 fixed positions are C are in blue.

G. Enrichment of HNRNPK motifs in eCLIP data. Linear: top 10 6-mers; Bipartite: top 10 spaced 6-mers; Degenerate: top 10 degenerate 12-mers with 6 Cs (those shown in **Fig. 5H** left). Error bars represent bootstrapped standard deviations (100 samples).

H, I. Enrichments for top 10 degenerate 12-mers with 6 Ns for CELF1 (**G**) and FUBP1 (**H**). Error bars represent bootstrapped standard deviations (100 samples).

Fig. S5

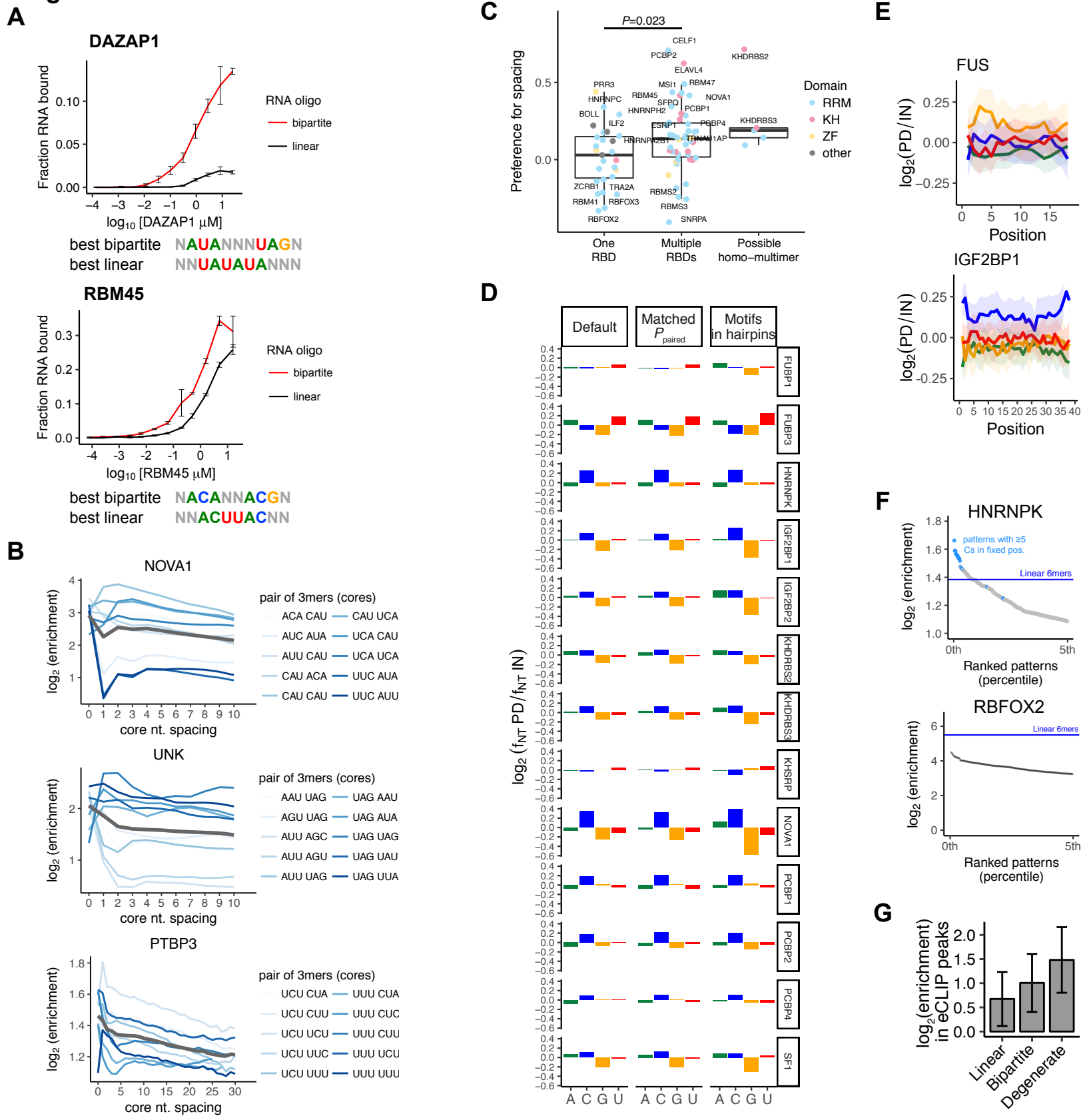
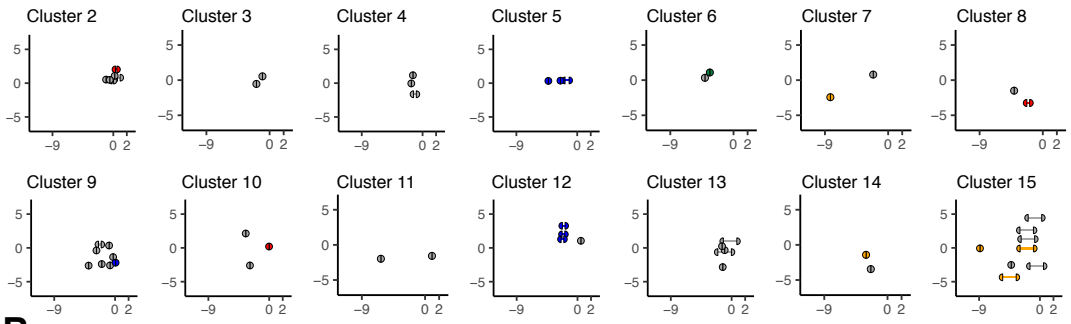

A
DAZAP1
Fraction RNA bound
log10 [DAZAP1 μM]
RNA oligo
bipartite
linear
best bipartite NAUANNNUAGN
best linear NNUAUAUANNN
RBM45
log10 [RBM45 μM]
best bipartite NACANNACGN
best linear NNACUUACNN
B
NOVA1
UNK
PTBP3
log2 (enrichment)
core nt. spacing
pair of 3mers (cores)
ACA CAU
AUC AUA
AUU CAU
CAU ACA
CAU CAU
CAU UCA
UCA CAU
UCA UCA
UUC AUA
UUC AUU
AAU UAG
AGU UAG
AUU AGC
AUU AGU
AUU UAG
UAG AAU
UAG AUA
UAG UAG
UAG UAU
UAG UUA
UCU CUA
UCU CUU
UCU UCU
UCU UUC
UCU UUU
UUU CUA
UUU CUC
UUU CUU
UUU UCU
UUU UUU
C
P=0.023
Preference for spacing
Domain
RRM
KH
ZF
other
One RBD
Multiple RBDs
Possible homo−multimer
D
Default
Matched P paired
Motifs in hairpins
log2 (fNT PD/fNT IN)
FUBP1
FUBP3
HNRNPK
IGF2BP1
IGF2BP2
KHDRBS2
KHDRBS3
KHSRP
NOVA1
PCBP1
PCBP2
PCBP4
SF1
E
FUS
IGF2BP1
log2(PD/IN)
Position
F
HNRNPK
patterns with ≥5 Cs in fixed pos.
Linear 6mers
log2 (enrichment)
Ranked patterns (percentile)
RBFOX2
G
log2(enrichment in eCLIP peaks
Linear
Bipartite
Degenerate
H
CELF1
degenerate
linear
log2 (enrichment)
patterns with Gs & Us in fixed positions
Linear 6mers
Ranked patterns (percentile)
I
FUBP1
patterns with ≥5 Us in fixed positions

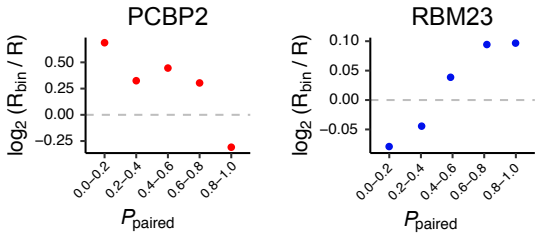
Figure S6. Sequence context effects on RBP binding, Related to Figure 6.
A. Dispersal of specificities as in **Fig. 6A** for all other clusters.
B. Example of the distance between feature-specific *R* value profiles for PCBP2 and RBM23 for P_{paired}.
C. Mean average P_{paired} over 6-mers in RBNS input 20-mers and human transcript regions (5' UTR, 3' UTR, CDS, Intron), separated by whether the 6-mer is a homopolymer ("1 base"), contains two distinct bases ("2 base"), or contains three or four distinct bases ("Complex").
D. Change in stem content ($=f_{stem,PD} \times \log_2(f_{stem,PD}/f_{stem,IN})$, where $f_{stem,PD}$ is shown for RBM22 and ZNF326 in **Fig. 4E,F**) averaged over the top 6-mer of each RBP, separated by RBPs that do vs. do not contain a ZF. *P*-value determined by Wilcoxon rank-sum test.
E. Comparison of differences in P_{paired} preferences among pairs of RBPs ($|P_{paired}\text{ ratio},RBP_1 - P_{paired}\text{ ratio},RBP_2|$, where P_{paired} ratio is $P_{paired,\ PD}/P_{paired,\ IN}$ averaged over the top 6-mer as shown in the right bar of **Fig. S4A**), separated by RBPs that do vs. do not contain a ZF. *P*-value determined by Wilcoxon rank-sum test.

Fig. S6

A



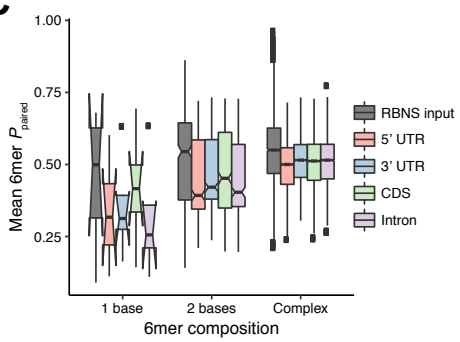
B



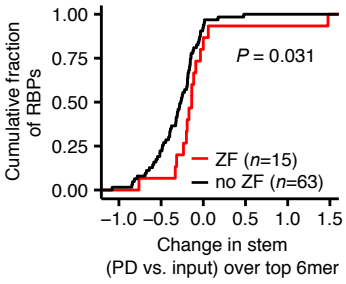
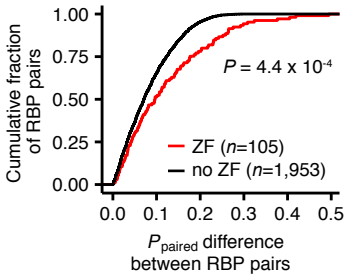
$\overrightarrow{\Delta R}_{PCBP2}$ = [0.69, 0.33, 0.44, 0.31, -0.31] $\overrightarrow{\Delta R}_{RBM23}$ = [-0.08, -0.04, 0.04, 0.10, 0.10]

Dist(PCBP2, RBM23) = 1.0 – corr($\overrightarrow{\Delta R}_{PCBP2}$, $\overrightarrow{\Delta R}_{RBM23}$) = 1.71

C



D



E